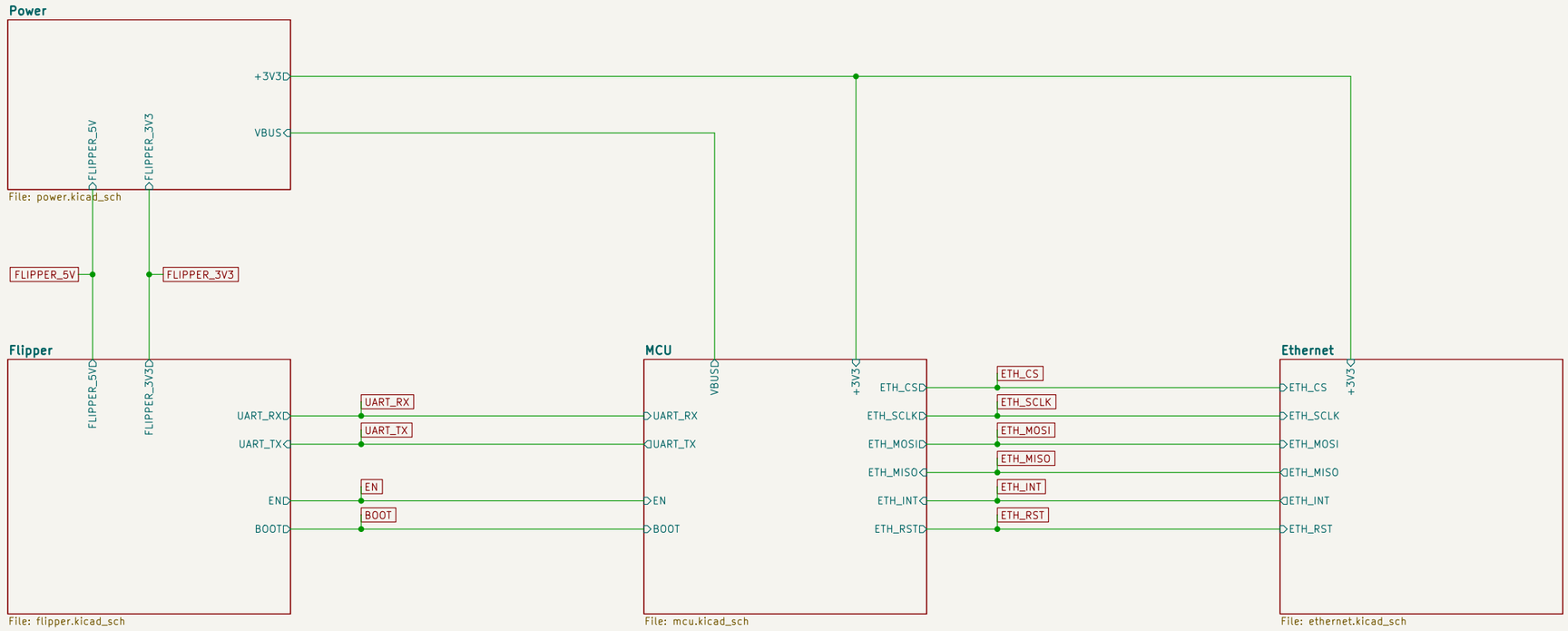
SSH Runner add-on for Flipper Zero — rev 0.1

ESP32-S3-WROOM-1-N16R8 + W5500 Ethernet + RJ45 magjack on Flipper Zero's GPIO header. Design notes and the reasoning behind every value: hardware/DESIGN.md.

Sheets

1 Overview (this sheet)
2 Power: input, LDO, power mux
3 MCU: ESP32-S3, USB-C, Qwiic, expansion
4 Ethernet: W5500, RJ45 magjack
5 Flipper: GPIO headers, UART



NOTES

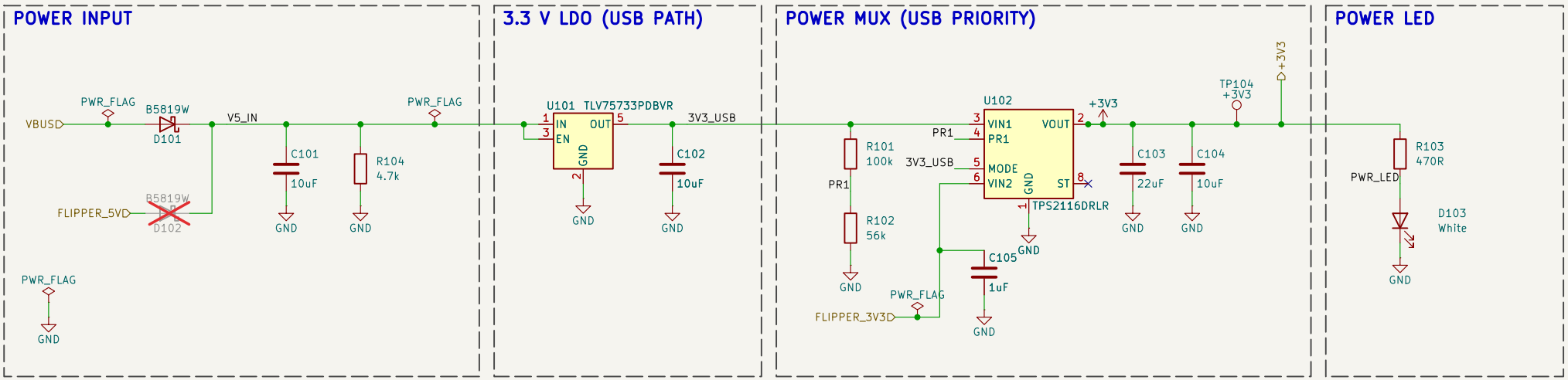
Supply: Flipper GPIO pin 9 (3.3 V) or USB-C 5 V through a 3.3 V LDO, ORed by a TPS2116 power mux with USB priority. All logic and I/O is 3.3 V.
Impedance (JLC04121H-7628 stack-up): Ethernet MDI pairs 100 ohm differential on F; USB D+/D- 90 ohm differential on In2 (DESIGN.md 5.5).
Not fitted (DNP): D102 (Flipper 5 V fallback), R403/R404 (Flipper reset/boot links), J202 (Qwiic connector), J203 (expansion pads).
D103 and D201 are reverse-mount LEDs on the parts side, shining through holes to the art side.
Board: 4-layer, 1.2 mm, 67.34 x 36 mm, assembled by JLCPCB (hardware/CAM.md). This drawing is generated by hardware/tools/gen_sch.py from the parts and nets it lists.

Design notes: hardware/DESIGN.md
License: CC-BY-NC-SA-4.0 (logo: all rights reserved, hardware/art/LICENSE.md)
SSH Runner add-on for Flipper Zero
flipper-ssh-runner

Sheet: /
File: ssh-runner.kicad_sch

Title: SSH Runner add-on for Flipper Zero

Size: A3	Date: 2026-10-25	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/5



NOTES

Two supplies, USB first. VIN1 = USB 5 V through D101 and the 3.3 V LDO U101; VIN2 = Flipper GPIO pin 9 (3.3 V). The TPS2116 picks VIN1 while PR1 (100k / 56k from VIN1) is above its 0.92–1.08 V reference, i.e. VIN1 above 2.56–3.01 V (at 3.3 V, PR1 = 1.18 V). MODE = VIN1: priority mode. It blocks reverse current, so USB never back-feeds the Flipper. Soft start 1.3 ms; about 74 uF on +3V3 (with 3V3A): about 190 mA inrush from the Flipper. ST (open drain) is unused.

D102 (DNP) is the 5 V fallback from Flipper pin 1: fit it and turn on '5V on GPIO' only if pin 9 proves too weak. R104 bleeds V5_IN: with USB unplugged, the ESP32-S3's D+ pull-up leaks through the USBLC6 into VBUS (about 2.7 V) and on through D101 and the LDO; 4.7k keeps PR1 well below 0.92 V, so the mux never picks the dead USB input.

TLV757P: 425 mV max dropout at 1 A; about 0.4 W at full load on USB (60–75 C rise, DESIGN.md 4.5). D103 is a reverse-mount LED: placed on the parts side, it shines through a hole to the art side. Bench test (DESIGN.md 4.6): prototype on Flipper pin 9 only, 10 min LAN + WiFi flood, no brownouts.

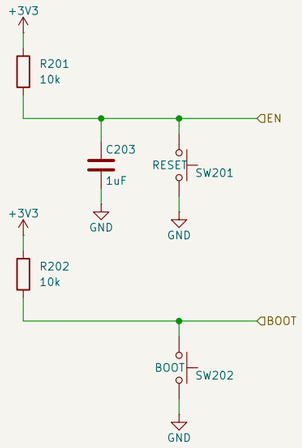
Design notes: hardware/DESIGN.md
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SSH Runner add-on for Flipper Zero
flipper-ssh-runner

Sheet: /Power/
File: power.kicad_sch

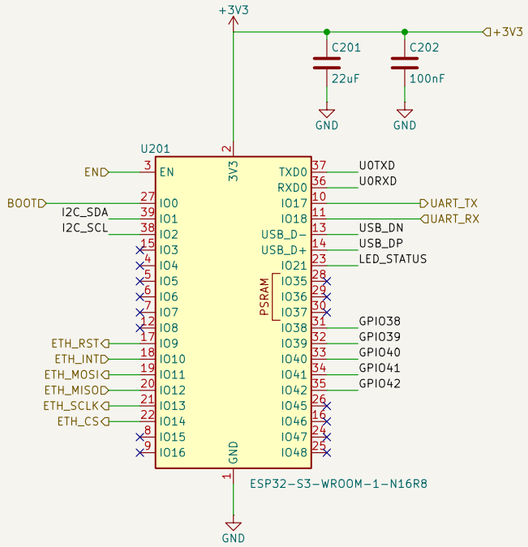
Title: Power: USB / Flipper input, LDO, power mux

Size: A4	Date: 2026-10-25	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 2/5

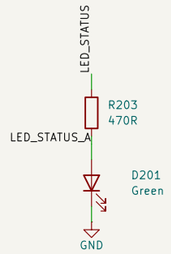
RESET / BOOT



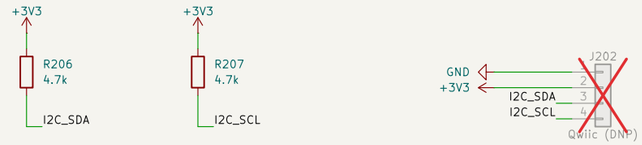
ESP32-S3-WROOM-1-N16R8



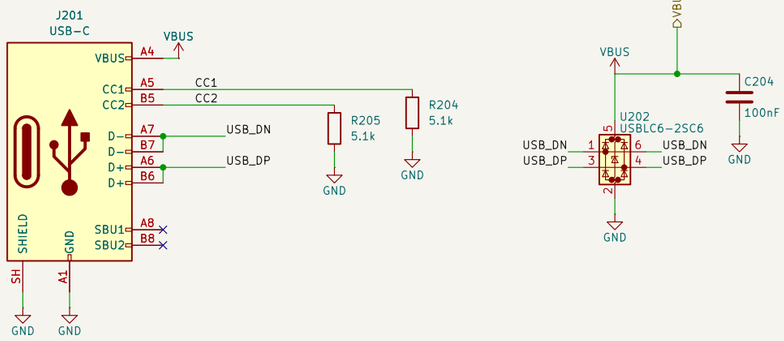
STATUS LED



QWIIC I2C (DNP CONNECTOR)



USB-C, CC, ESD

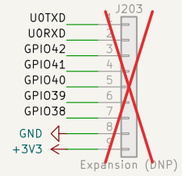


NOTES

GPIO map = the prototype's defaults (firmware-esp32 Kconfig):
- W5500: SCLK IO13, MOSI IO11, MISO IO12, CS IO14, INT IO10, RST IO9
- UART1 to the Flipper: TX IO17, RX IO18 (1k series resistors on the Flipper sheet)
- USB IO19/IO20: USB-Serial-JTAG console, flashing
- UART0 IO43/IO44 on the expansion pads J203 only (ROM boot log, never to the Flipper)
- Status LED IO21; Qwiic SDA IO1, SCL IO2; spare IO38-IO42 on J203

All I/O is 3.3 V (ESP32-S3, W5500, Flipper GPIO). Strapping pins IO3/IO45/IO46 and the octal-PSRAM pins IO35-IO37 are left open.
EN: 10k / 1uF RC per Espressif.
USB D+/D-: 90 ohm differential pair (0.23 mm / 0.22 mm on In2); ESD array next to the receptacle.
D201 is a reverse-mount LED, shining through a hole to the art side.
DNP: J202 (Qwiic connector), J203 (expansion pads).

EXPANSION / UART0 PADS (DNP)



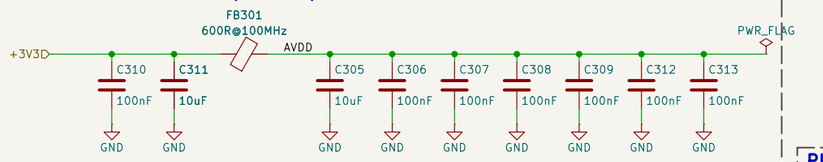
Design notes: hardware/DESIGN.md
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SSH Runner add-on for Flipper Zero
flipper-ssh-runner

Sheet: /MCU/
File: mcu.kicad_sch

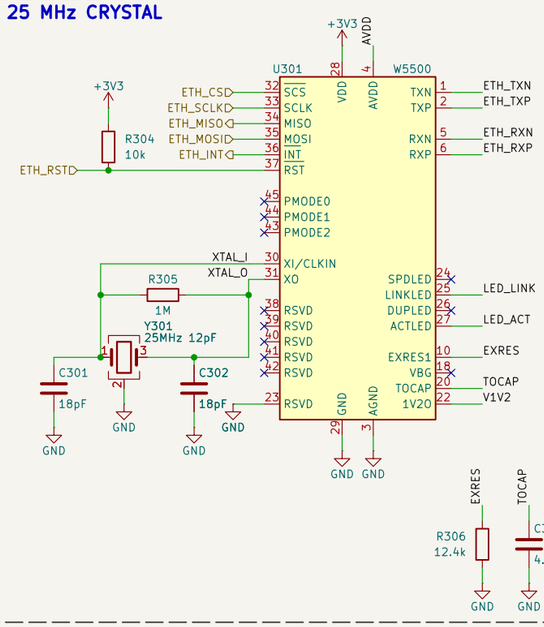
Title: ESP32-S3 module, USB-C, Qwiic and expansion

Size: A3	Date: 2026-10-25	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 3/5

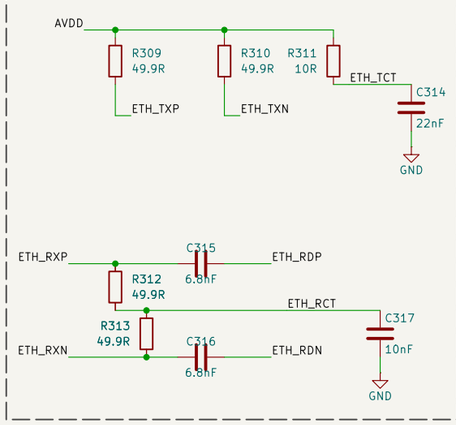
SUPPLY: VDD 3V3, AVDD (FERRITE)



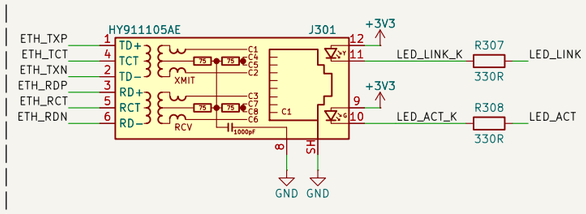
W5500, 25 MHz CRYSTAL



PHY TERMINATION (WIZnet REFERENCE)



RJ45 MAGJACK HY911105AE, LINK/ACT LEDs



NOTES

W5500 per datasheet v1.0.5: EXRES1 12.4k 1% to GND, TOCAP 4.7uF, 1V2O 10nF, VBG open, RSVD pin 23 to GND,
PMODE[2:0] open (internal pull-ups: auto-negotiation, all capable). No auto-MDIX in the W5500.
Crystal C9006 is a 12 pF part: load caps 2 x (12 - ~3 pF stray) = 18 pF.
SPI direct (no series resistors), 20 MHz, all 3.3 V.

MDI pairs ETH_TX/ETH_RX: 100 ohm differential on F over In1 (0.25 mm at the chip's 0.5 mm pitch, 0.23 mm
on the 0.45 mm TX run, matched within 0.3 mm), no other signals between; DESIGN.md 5.5.
Jack: HY911105AE (HR911105A footprint), -40 to 85 C, pin 8 = chassis cap (to GND).
Jack LEDs as in WIZnet's reference: yellow (12/11) = LINK, green (9/10) = ACT; anodes to 3V3,
cathodes through 330R to the W5500's LINKLED/ACTLED (active low).

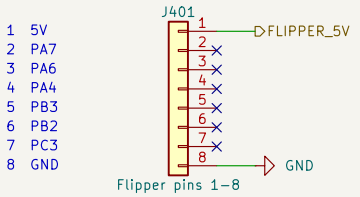
Design notes: hardware/DESIGN.md
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SSH Runner add-on for Flipper Zero
flipper-ssh-runner

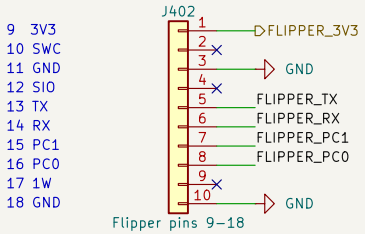
Sheet: /Ethernet/
File: ethernet.kicad_sch

Title: W5500 Ethernet controller, RJ45 magjack

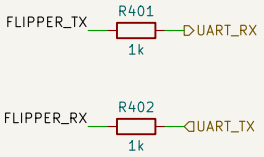
Size: A3	Date: 2026-10-25	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 4/5

FLIPPER ZERO GPIO HEADERS



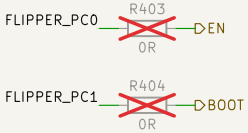


UART1, 1k SERIES



921600 baud, 3.3 V. The resistors limit the current when one side is unpowered.

RESET / BOOT LINKS (DNP)



NOTES

Two right-angle 2.54 mm headers (8 + 10 pins) at the positions of Flipper's official external-module drawing (hardware/ref/flipper-3d-models): pin n of J401 is Flipper pin n, pin n of J402 is Flipper pin n + 8.
Flipper GPIO is 3.3 V. Pin 9 (3V3) is the board's main supply; pin 1 (5V) is used only with D102 fitted.
R403/R404 (DNP): pin 16 (PC0) to EN, pin 15 (PC1) to GPIO0, for a future app that resets the ESP32 or enters download mode; the app must drive them open-drain.

Design notes: hardware/DESIGN.md
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SSH Runner add-on for Flipper Zero
flipper-ssh-runner

Sheet: /Flipper/
File: flipper.kicad_sch

Title: Flipper Zero GPIO headers

Size: A4	Date: 2026-10-25	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 5/5